MINIFLY V212 / EXPERIMENT DESIGN / 01

Learn what an event changes

Can one local learner acquire a transition rule from observed consequences, preserve unrelated state, and execute the rule on new entities and event sequences? V212 removes language and answer retrieval from this test. It keeps structured perception as a declared experimental scaffold.

Why this test

In the reviewed V211C direct-memory path, a same-person object event reduced location accuracy to 0/256 under both vocabulary conditions. The correct location disappeared from 497 of 512 candidate sets. The replacement writer overwrites shared entity rows. This diagnoses the tested direct path. Round 17’s exact learner and receipts were unavailable for that audit.

What the learner observes

The stream contains event boundaries, stable opaque entity IDs, opaque action IDs, typed argument slots, and intermittent sensor observations of an entity at one of four locations. Entity IDs select working-state slots and have no trainable identity embeddings. No action names, latent relation labels, holder fields, support indexes, or flags indicating which state should change reach the learner.

At a sensor request the learner gets only the entity ID. It commits a distribution over locations before seeing the observation. During training, the revealed position provides prediction error and then becomes a generic observed input. During frozen evaluation, reveal only prescheduled context observations. A scored probe withholds its value from the learner for the rest of the trajectory.

Same interface, two different dynamics

Use matched simulator families with the same observable fields and opaque event vocabulary. A contact-like event establishes a lasting dependency in one family and copies a location once in the other. A release-like event ends the dependency where present. The simulator alone applies these semantics. Both families include independent motion and coincidental co-location. Train a separate parameter instance per family using identical architecture, initialization and budgets. No family ID enters the learner.

Use paired histories with the same last observation and next event, but different intervening contact or release. Include reattachment, an unrelated entity event, and an actual change that invalidates an earlier observation. Use the same action-code mapping within each train-to-test split, and vary it across paired worlds. Renaming action symbols alone is not the test of a different dynamic.

An executable local learning rule

Forward computation

Start with up to six entities, eight stochastic binary units per entity, and four per ordered pair of distinct entities. A shared entity cell reads its own prior state, the fixed mean of incoming pair states, and the observed event code and argument indicators. A shared pair cell reads its own prior state, the two endpoint states, and the event code and indicators. Both use the previous state in synchronous updates. Observation tokens use the same input interface. Pair aggregation has fixed weights and uses no attention.

For each cell j, form local features x from those inputs, compute p = sigmoid(w dot x + b), and sample h from Bernoulli(p). Bias uses a constant input. Read sensor probabilities from the queried entity state with a shared linear four-location softmax head. No action-conditioned branch performs copying, attachment, detachment or movement. No hard-coded state table provides a location prediction.

How error reaches transition weights

On every transition, each weight accumulates a score trace e += (h - p) x. Clear traces at episode start. At each revealed training observation, compute loss L = -log p(correct location). With baseline B from earlier episodes at the same observation index, accumulate the local transition update -eta (L - B) e. Keep the trace through later observations, and apply accumulated updates only after the episode. Keep the baseline fixed within the episode, then update it with a running average.

This is a sampled score-function update. The cell supplies its own eligibility, and the sensor loss supplies a scalar broadcast. Holding parameters fixed within each short episode keeps the accumulated trace tied to the sampled transitions. The readout separately receives the usual local softmax delta using the revealed location and the queried state. Gradients never propagate through recurrent states. Save a matched no-transition-write arm with the same trainable readout.

Check credit before a full run

On short engineering trajectories, compare the mean sampled update with a finite-difference estimate on a tiny cell fixture. Verify that transition weights change and frozen-transition weights do not. Compare subsequent loss with matched sampling draws and a readout-held-fixed diagnostic. Treat persistent failure to produce a useful transition signal as a qualification failure. Report episode length, trace memory, sampling variance and update norms.

Test execution, persistence, and transfer

Paired evaluation

Propose 12 paired development-to-evaluation worlds per family. Seal short teaching episodes and test trajectories with new IDs and untaught event compositions. Keep episode length bounded at 24 event/observation tokens for the initial test. Reset transient state at each new-world boundary. In frozen evaluation, update only inferred state. Average 32 fixed-seed state trajectories per forecast, and apply the same inference budget across model arms.

Score paired histories separately when the same last observation and current event imply different consequences. Report changed-state accuracy, still-valid accuracy after an unrelated event, and sensor bits per observation. Include delayed queries, for which the target entity has not just been observed. Report each family separately.

Controls

Compare last observation, a frozen-transition learner with trainable output, and the full model. In a transition-credit shuffle control, keep inputs and readout learning intact but permute centered loss signals across matched episodes at the same observation index. Match the update count and keep true observations available only at the original assimilation times. Use a simulator-side reference to verify that the supplied history determines the scored outcome. The reference does not supply hidden state to MiniFly.

Proposed gates and forward transfer

At matched training dose, require at least 85% frozen accuracy on changed-state probes and 90% on still-valid probes in each family. Require at least +10 percentage points on changed-state probes versus both frozen-transition and shuffled-credit controls, with paired 95% intervals excluding zero. Report last-observation and reference results even if the gate fails.

Separately compare an experienced learner with a fresh one on new worlds of the same family. Reset state and eligibility in both; retain only learned weights in the experienced arm and start fresh in both arms' loss baselines. Match downstream observations and updates. Report pre-update prediction loss on the first 64 revealed observations, and require at least 0.05 bits per observation mean improvement with a paired interval excluding zero for a forward-transfer claim.

Execution and scope

Qualify causal scoring and label isolation, local credit, and fresh-process checkpoint recovery before opening evaluation worlds. Freeze the roster, thresholds, rates, random seeds and resource limits on engineering worlds. A pass demonstrates bounded learned dynamics under structured perception. Language, learned perception and autonomous action are separate steps. Basis: the user-provided V211B/V211C audit and Round 18 proposal. V212 is a design; no runs were launched.